2025 IEEE International Conference on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON)29-30 November 2025, Eastern University, Dhaka, Bangladesh

2025 IEEE International Conference on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON) | 979-8-3315-6105-5/25/$31.00 ©2025 IEEE | DOI: 10.1109/BECITHCON69222.2025.11503942

Obesity Prediction with Optimized Feature Selection and Ensemble Learning Techniques

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***Abstract*— Obesity is a multifactorial public health challenge linked to chronic diseases such as diabetes, cardiovascular disorders, and cancer. The detection of risks of obesity should be done at an early stage. Nonetheless, conventional approaches may not be able to capture these influences being nonlinear and multispectral. The development of a machine learning model that uses optimized feature selection, dimensionality reduction and ensemble learning for the classification of obesity is discussed in this paper. The data was collected through an online survey conducted with 1610 respondents in Turkey. SMOTE, SelectKBest, and PCA were performed to preprocess the data for balanced classes. Out of the thirteen machine learning classifiers that were assessed, Decision Tree, Random Forest and XGBoost were the best. The stability-predicting models were designed using voting ensembles to further enhance the predictive stability of the models when put together. When all the models were combined, the hard voting technique achieved the highest accuracy of 92%, which is higher than that of all other models and previous benchmarks. The usefulness of combining preprocessing and employing an ensemble for obesity prediction models was demonstrated. We intend to utilize explainable AI techniques in upcoming research like SHAP and LIME. These methods enhance expressiveness and clinical trust, which may enhance the use of machine learning in personalized preventive healthcare.**

Keywords*— Obesity Prediction, Machine Learning, Feature Selection, Ensemble Learning, SMOTE, PCA.

I. INTRODUCTION

Today, obesity is one of the most serious health problems, which affect more than a million people all over the world. It is closely linked with early death and a variety of non-communicable illnesses such as type 2 diabetes, heart diseases, and some cancers [1]. According to the global studies, obesity prevalence is rising steadily in developed and developing countries alike. Moreover, the rise is sharp in less developed countries. The body weight, body mass index (BMI), and muscle size have conventionally been used as measures of obesity because of their simplicity. But these indicators do not always give the complete picture of someone's health status. To overcome these limitations, researchers have been increasingly using machine learning (ML) techniques to find the causes of obesity and to predict its occurrence even before it occurs [2], [3]. In this area, random forest and extra trees have shown a great potential to predict. Ensemble learning with SMOTE as a resampling method gained prediction accuracy more than 90% and AUC value near to 98% [4]. Hybrid metaheuristic approaches which combine neural networks with optimization techniques have shown accuracies up to 92% and also enhance the transparency of decision-making. Recent developments indicate that not only do ML techniques enhance predictive performance, but they also deal with obesity complexity.

Another critical progress in this area is model interpretability. Interpretability is as important as accuracy in medical and clinical applications because a healthcare professional needs to understand the reason for a model's prediction.

Obesity risk factors include waist-to-hip ratio and physical activities, diet patterns, and self management behaviors, which have been explained using Shapley values (SHAP) techniques. Such transparency enables practitioners to translate model outputs into actionable and personalized recommendations for their patients [5]. The obesity-related datasets always contain an imbalance (around two-thirds of archives are unhealthy). Predictive reliability may be low because minority categories such as severe obesity or underweight are rarely used. Let's look at the problem in this paper. Then, we will talk about the solution to the problem. These methods make models stronger, especially for small classes in clinical studies [6][7]. Medical machine learning research shows that managing class imbalance is key to achieving fairness and improving applicability in practice [8]. Previous research on datasets from specific countries adds support for the value of ML. The results suggest that simple ML models do well. Also, including recent preprocessing and ensembles can improve the results even more.

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These findings encourage a comprehensive ML framework for obesity prediction which is proposed by this study. The results obtained by using the proposed method of synthetic oversampling, feature selection, dimensionality reduction using PCA and final classification using ensemble voting methods result in more accurate and generalizable results on diverse datasets. The framework intends not only to enhance prediction performance in healthcare but also to discourage negative or unwanted artificial intelligence (AI) developments by focusing on emerging trends in medical AI research.

Overall, the research contributes the following:

- collected data from an online survey with 1,610 people who lived in Turkey.
- Using techniques such as SMOTE, SelectKBest and PCA.
- Evaluating 13 machine learning classifiers for the prediction of Obesity.
- Using hard and soft voting in ensemble learning.
- Used Accuracy, Precision, Recall, F1-Score and ROC as metrics to evaluate their models.

II. Literature Review

Recent studies show that machine learning (ML) significantly helps predict and classify obesity using the interplay of demographics, behaviors, and lifestyle factors. Compared to earlier statistical methods, ML models in clinical settings are capable of improving accuracy while being interpretable.

Fujihara et al. [9] built a machine learning framework to predict body weight of adults using large-scale public health datasets. Their work has demonstrated that multiple algorithms can be assessed not just for accuracy but also for interpretability in health managements. Zhou et al. [10] presented mini-review paper in which they described the various applications of ML in research on obesity in the region of nutrition, behavior and environment. In their further research, Sun et al. applied interpretable models based on Shapley values to pooled national surveys. Their study found that sedentary habits, alcohol intake, and protein consumption were major predictors of obesity [11]. In a systematic review based on PRISMA standards, Ferrers et al. [12] found that deep learning can help in learning complex features. However, they also found that traditional machine learning methods like Random Forest, SVM and Logistic Regression, which are relatively transparent and interpretable, dominate obesity prediction papers. Many obesity studies find that their dataset suffers from imbalanced classes. For instance, the other obesity categories such as severe or morbid obesity have little data to contribute, which can cause problems in the training process of the forecasting tool. Researchers have combined ensemble classifiers with oversampling techniques to deal this. According to a survey conducted by Rawat et al. [13] who surveyed the various oversampling techniques such as SMOTE and its adaptive variants. As per them, the predictive power is considerably enhanced when they are used with ensemble tree-based models in imbalanced settings. Salmi et al. [14] found that through synthetic oversampling, Random Forest, Extra Trees, and Gradient Boosting algorithms constructed robust models that improved AUC, stability, and fairness.

The Turkish lifestyle dataset is a benchmark dataset for obesity prediction. Koklu and Sulak [15] used various AI models such as Random Forest, ANN, k-NN, and SVM on their 1610 individuals and concluded that Random Forest was superior to other concurrent approaches with 87.82% success. To build on this work, Arslan and Yapici [16] proposed a hybrid ensemble using Extra Trees, Decision Tree, MLP, Random Forest and SVM boosted with SMOTE resampling and achieved an accuracy of 91%. As a result, hybrid models that contain both feature diversity and balancing advantage had the best performance. Earlier works also contributed foundational insights. Dugan et al. [17] achieved 85% accuracy using Naive Bayes to predict early-life obesity. Hammond et al. [18] applied LASSO regression to electronic health record data, achieving an accuracy of 81.8% for girls and 76.1% for boys. According to Dunstan et al. [19], the Random Forest was used for nationwide obesity prediction using food sales, which performed unevenly across countries. While they had limitations, these studies paved the way for future enhancements with better ML methods.

One long term issue faced by ML in health care is interpretability. Jiang et al. [20] made use of LightGBM along with SHAP to predict social isolation in older adults to signal the potential of interpretable ML in revealing hidden social determinants of health. Diabetes, depression and obesity research have also been studied with similar techniques. For example, Liu et al. [21] implemented XGBoost with SHAP on a Chinese dataset of 5,482 adults. This implementation showed an overall accuracy of 83% however it performed with lower effectiveness (~61.8%) for the case of Class II obesity. Thus, indicating that this algorithm causes a trade-off between its accuracy and fairness with respect to classification of minority categories.

Of late studies have investigated the methods deep learning and boosting-based. Hasan et al. [22] proposed a deep neural network to predict obesity in adolescents. Their approach achieved 88.42% accuracy on a small health screening dataset. Hossain et al. [23] studied the gradient boosting and ensemble models on the Spanish obesity data (n = 1057) in a different study. They reported a maximum accuracy of 79%. Although it is lower than the Turkish datasets. It shows that the use of the Turkish dataset does not ensure success in different nationalities. Existing studies handle class imbalance, feature selection, or ensembles separately, and models on the Turkish lifestyle dataset still show limited accuracy and stability. No prior work combines SMOTE, SelectKBest, PCA, and a hard-voting ensemble within a single optimized pipeline. As a result, the proposed framework achieves higher predictive performance than existing methods on the same dataset.

III. Proposed Methodology

For our proposed obesity prediction framework, we have used a dataset collected from an online survey conducted in Turkey [24]. The objective is to help healthcare professionals to identify different obesity levels using data like age, eating, usages and physical activity. To understand data distribution, detect correlations, and ensure data quality the process starts with exploratory data analysis (EDA). Next is the

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preprocessing phase, where 80% of the data is allocated for training and 20% for testing. Then we scaled the dataset using standardization and balanced class imbalance using SMOTE. After preprocessing, feature engineering has been performed that involves selecting significant features using SelectKbest and applying Principal Component Analysis (PCA) for dimensionality reduction. We trained 13 different ML classifiers and evaluated their performance. The top-achieving models are then combined using a Voting Classifier to elevate accuracy and robustness. Final evaluation is done using metrics such as accuracy, precision, recall, F1-score, and ROC AUC to identify the best model.

The full architecture of our proposed obesity prediction system is depicted in "Fig. 1".

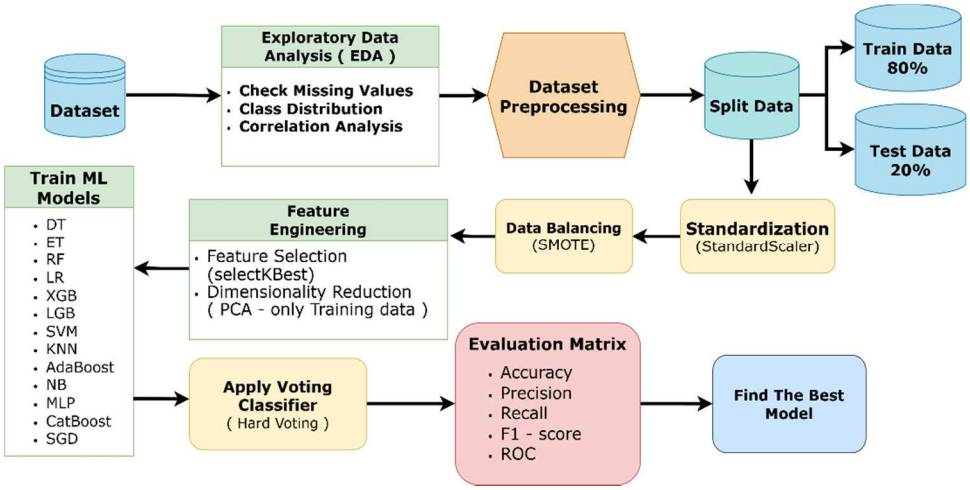


Fig 1: Obesity Prediction Model

A. Dataset

The dataset used, sourced from an online survey conducted in Turkey with 1,610 participants, including 14 variables related to health and lifestyle such as age, physical activity, fast food consumption, and dietary habits. Key attributes include age, sex, height, weight, fast food consumption, physical activity, and calorie intake, among others. The target variable is the obesity class, helping assess how various behavioral and biological factors influence obesity.

B. Exploratory Data Analysis (EDA)

To perceive the dataset structure and quality, we checked for missing values (none found). And all categorical features are already encoded into numeric values. Examined class distribution to assess imbalance across obesity categories. Performed correlation analysis using a heatmap to detect multicollinearity and select relevant features.

C. Data Preprocessing:

Data preprocessing transforms raw inputs into a format suitable for ML. The following steps were applied to enhance model performance and generalization:

1) Feature Scaling

Since some features are numerical, StandardScaler was used to normalize them by removing the mean and scaling to unit variance:

$$X' = \frac{X - \mu}{\sigma} \qquad (1)$$

Where X represents the original value, μ is the mean, and σ is the standard deviation of the feature. This standardization ensures that each feature has an equal influence.

2) Class Balancing with SMOTE

To correct class imbalance, Synthetic Minority Oversampling Technique (SMOTE) was applied on the training set. SMOTE creates synthetic samples for minority classes by interpolating between existing samples and their nearest neighbors, improving classifier sensitivity across all classes.

3) Feature Selection using SelectKBest

To retain only the most relevant features, we used SelectKBest, which selects the top k features based on statistical tests like ANOVA F-value or mutual information:

$$Score(X_i) = f(X_i, Y) \qquad (2)$$

where X_i is the feature and Y is the target. Features with the highest scores were retained, reducing noise and improving learning.

In our dataset, out of 14 input variables, 9 features were selected as the most significant: Sex, Age, Consumption of Fast Food, Frequency of Consuming Vegetables, Number of Main Meals Daily, Smoking, Calculation of Calorie Intake, Physical Exercise, and Type of Transportation Used. These features demonstrated the strongest statistical relationship with the target obesity classes, making certain that the model trained on the most explanatory variables.

4) Dimensionality Reduction using PCA

To make the data cleaner and more efficient, PCA was performed on the standardized training data, diminishing it to a smaller set of dimensions (Z = XW) while keeping most of the variance. This not only sped up the training process but also reduced overfitting, ensuring the data was balanced and properly scaled for robust model building.

In our study, PCA was applied to the nine features selected by SelectKBest, lowering the dimensionality while keeping 95% of the original variance. This process condensed the feature space into a reduced set of independent principal components, giving us a cleaner and more compact

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representation without losing any key information.
These preprocessing steps collectively make sure that the input data was balanced, scaled, relevant, and efficient, laying the foundation for robust ML model development.

D. Machine Learning Models

In this study, we gave careful consideration to bias and variance when selecting ML models for obesity prediction. We have implemented 13 different ML classifiers. These include Decision Tree, Extra Tree, Random Forest, XGBoost, LightGBM, SVM, K-Nearest Neighbors (KNN), AdaBoost, Logistic Regression (LR), Naive Bayes (NB), Multilayer Perceptron (MLP), CatBoost, and Stochastic Gradient Descent (SGD) classifier.
An ensemble learning technique, the Voting Classifier applied to enhance prediction accuracy and reduce both bias and variance. Ensemble learning combines predictions from several models to provide better predictions than any one model alone. It leverages the advantages of several models to enhance classification accuracy, robustness, and generalization capacity.
In our procedure we utilized both soft voting and hard voting approaches with the best performing models: DT, RF and XGBoost. After evaluation, hard voting exhibited the most accurate prediction performance and also had the highest accuracy vs. soft voting and any of the individual models on their own. The use of ensemble methods allowed for interpretability and stability of the DT and RF models to be combined with the predictive capacity of the gradient-boosting models, XGBoost. The outcome was a more accurate and reliable obesity classification system that has been developed based on this combination of models.
The hyperparameter settings used for these top models are summarized in Table I.

TABLE I. HYPERPARAMETER VALUES FOR EACH CLASSIFIER

Classifier	Parameter	Value
Decision Tree	Criterion	gini
	Max_depth	None
	Random state	5
Random Forest	n_estimators	100
	Max_depth	None
	Random state	42
XGBoost	n_estimators	100
	Learning_rate	0.3
	Max_depth	6
lightGBM	n_estimators	100
	Learning_rate	0.1
	Random state	42
Voting Classifier	Voting type	Hard

The findings of this study indicate that ensemble learning, which includes a hard voting process, can enhance the accuracy of obesity prediction. This resource would enable healthcare providers to have an accurate, consistent resource to guide their initial effort of obesity identification and precise classification to promote intervention or treatment based on the risk level of obesity.

E. Performance Metrics

In order to assess how well our obesity classification models performed, we calculated key evaluation metrics, including accuracy, precision, recall, F1-score and ROC AUC. With these metrices we can assess the effectiveness of the models. As indicated by Eq. (3), Accuracy assesses the overall correctness of the model. Precision, based on Equation (4) measures out of all predicted positives how many are actually positive. The F1-score, given in Eq. (5), is the harmonic mean of precision and recall. Recall, shown in Eq. (6), evaluates how the model is competent enough to classify all actual obese cases. ROC AUC shows how well the model separates classes by plotting true vs. false positives. We also used a confusion matrix to analyze forecasts as True Positives (TP), False Positives (FP), and so on, giving us a clearer picture of performance.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

$$Precision = \frac{TP}{TP + FP} \quad (4)$$

$$F1 - Score = \frac{2 \times TP}{2 \times TP + FP + FN} \quad (5)$$

$$Recall = \frac{TP}{TP + FN} \quad (6)$$

IV. RESULT ANALYSIS

The confirmed findings were obtained from evaluating the proposed obesity prediction models across various machine learning algorithms. A total of 13 models were implemented and assessed, including DT, ET, RF, XGB, LGB, CatBoost, SVM, KNN, AdaBoost, LR, NB, MLP, and SGD Classifier. The proficiency of these models was compared adopting evaluation metrics such as Accuracy, Precision, Recall, and F1-Score. Table II presents a comparative summary of selected top-performing models based on their predictive performance.

TABLE II. EVALUATE DIFFERENT TECHNIQUE OF ML

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
XGBoost	91.30	91.55	91.30	91.36
Random Forest	90.99	91.06	90.99	90.98
Decision Tree	90.99	90.99	90.99	90.99
LightGBM	89.75	89.94	89.75	89.72
CatBoost	88.82	89.25	88.82	88.85
Extra Tree	86.65	87.12	86.65	86.65
Voting Classifier	92.00	92.00	92.00	92.00

As evident in Table II. The top-performing individual classifiers RF, XGB, and DT were chosen based on their high accuracy, interpretability, and efficiency in handling tabular

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data. RF is a popular ensemble method that assembles multiple DT and uses majority voting to upgrade accuracy and reduce overfitting. XGBoost is a powerful boosting algorithm prominent for its efficiency and strong predictive ability on structured data. Decision Tree is a simple yet effective classifier that splits data based on feature values to make predictions. These models consistently performed well, so we combined them using both hard and soft voting to boost overall stability and accuracy. Among these, hard voting gave the most accurate results, reducing variance and improving performance. The results of the ensemble approach are detailed below.

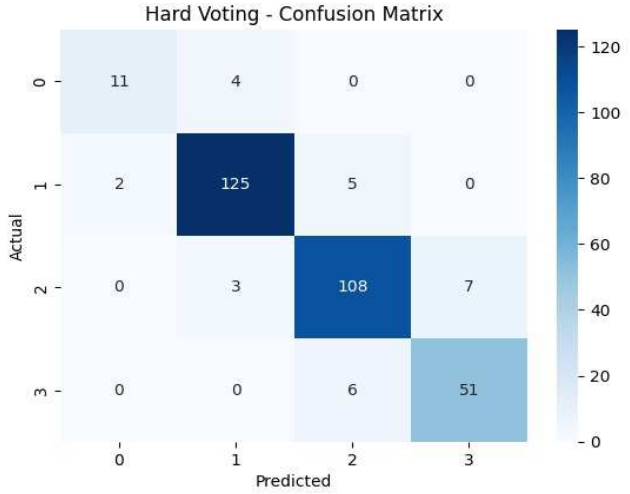


Fig 2: Confusion Metrics of DT, RF, XGBoost with voting.

The confusion matrices highlight the classification performance of these ensembles, showcasing improved accuracy and reduced misclassification. Additionally, ROC curves were generated to compare classifiers based on their discriminatory ability.

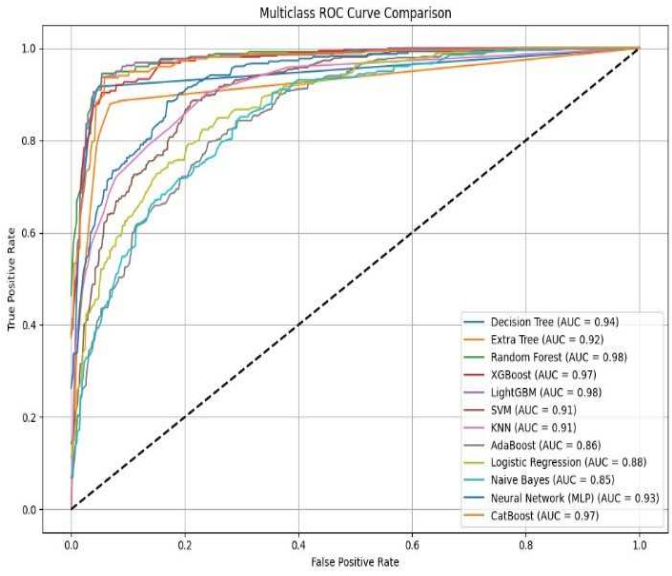


Fig 3: ROC curves of different ML classifiers

Ultimately, the results show that Random Forest, XGBoost, and Decision Tree achieved the most favorable and precise accuracy. Among these, Random Forest achieved 90.99%, XGBoost 91.30%, and Decision Tree 90.99% accuracy. With hard voting, the three models produced a highly desirable accuracy of 92%, with greater precision and accuracy. They compose a clearly compelling ensemble model in overall prediction performance for classifying obesity.

TABLE III: EVALUATING MULTIPLE ENSEMBLE MODEL

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
DT, RF	90.99	91,00	91.00	91.00
DT, XGB, LGB	90.99	91.00	91.00	91.00
RF, XGB, LGB	90.98	91.00	91.00	91.00
DT, RF, XGB	92.00	92.00	92.00	92.00

TABLE IV: COMPARISON OF THE ACCURACY OF OBESITY CLASSIFICATION BETWEEN THE PROPOSED METHOD AND EXISTING WORKS

Authors	Data Modality	Dataset Used	No. of Instances	Techniques Used	Remarks / Accuracy
Koklu and Sulak (2020)	Demographic and Lifestyle	Turkish Population [25]	1610	RF, SVM, DT, ANN	RF: 87.82%, no ensemble or balancing
Hossain et al. (2024)	Clinical and Lifestyle	Spanish Obesity Data	1057	XGBoost, RF, LR	79%
Liu et al. (2024)	Clinical	Chinese adult dataset	5482	XGBoost + SHAP	Class II obesity prediction: 61.76%, others ~83%
Arslan and Yapici (2025)	Demographic and Lifestyle	Turkish Population [25]	1610	ETC, DT, MLP, RF, SVM (with SMOTE)	91% (ETC + SMOTE)
Hasan et al. (2023)	Adolescent Obesity Prediction	Health Screening	321	Deep Neural Network	Overall Accuracy: 88%
Proposed Model	Clinical and Lifestyle	Turkish Population [25]	1610	SMOTE + SelectKBest + PCA + Voting	92% Accuracy improved

V. CONCLUSION

Timely and accurate prediction of obesity is important for effective intervention and sustained health. We proposed a predictive model using ensemble methods of machine learning, that is simultaneously comprehensive and high-performing. We proposed a solution that combines optimized feature selection (SelectKBest), dimensionality reduction

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(PCA), and class balancing (SMOTE). By improving the robustness and the accuracy for classifying obesity status, we help mitigate bias and variability in assessing obesity levels. The accuracy improved to an outstanding 92%, based on models derived through ensemble techniques such as hard voting, which exceeded previous benchmarks. In our evaluations, we explored algorithms including Random Forest, Decision Tree, XGBoost, and LightGBM, thereby also incorporating more than one model to mitigate overfitting and support generalization. These outcomes reinforce the promise of applying machine learning to enhance predictive analytics for potential obesity-related health risks.

To achieve greater trust and adoption into clinical practice, the embedding of Explainable AI (XAI) methods including SHAP and LIME are critical. These techniques can identify feature contributions and create an environment in which the model's decisions may transfer to healthcare professionals. Future efforts will integrate these explainability methods to foster transparency of the model, facilitate ethical implementation of the model, and ultimately support the medical decision-making process. This study demonstrates the potential of committee-based modelling approaches through data preprocessing, feature engineering, and ensemble methods to combat complex multiple modality diseases, such as obesity, while offering personalized and preventive solution possibilities.

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